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An Analytical Approach to Enhancing Recovery by Multi-Scale Geomechanics Reconstruction in Unconventional Resource

Jianhai Li, Fores, Chengdu, Technology Co., Ltd, Chengdu, Sichuan, China; Shenglong Wang, Guizhou Shale Gas Exploration and Development Co, Ltd, Gas Extraction Branch, Zunyi, China; Dong Huang and Yu Liang, Fores, Chengdu, Technology Co., Ltd, Chengdu, Sichuan, China; Xingning Huang and Xi Yan, CK Energy Technology Co., Ltd, Chengdu, China

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Abstract

As a kind of technological treatment that can be adopted to rapidly increase the well productivity hydraulic fracturing has been widely used in the development of unconventional resources around the world. Because the unreasonable multi-stage fracturing design will cause insufficient utilization efficiency of the resources and significant increased operation cost. Therefore, operators urgently need the technology for determining the optimal length of the multi-stage fracturing with different geological characteristics. Tight gas represents a critical unconventional resource, accounting for over one-third of China's natural gas reserves with substantial potential-particularly in Sichuan Basin's favorable zones holding nearly 7×10^{12} m³ resources at only 12% recovery. While replicable high-productivity well patterns in the Shaximiao Formation achieve record test rates, sustainable scale development faces technical constraints. Geological advantages include abundant resources, favorable accumulation conditions, and established core technologies with sustained cost reductions. However, challenges persist in channel sand characterization, seismic sweet-spot prediction, and drilling efficiency. This study optimizes stimulation design and parameters through geomechanical analysis, implementing geoengineering-integrated modeling to enhance reservoir conductivity, maximize channel sand utilization, and ensure sustained well productivity.

Geomechanical modeling

Based on rock mechanics log interpretation results, vertical mechanical property analysis was conducted for the target Sha-1 Member reservoir in Well Jinqian 201. The elastic modulus profile indicates static Young's modulus ranging between 15-25 GPa and static Poisson's ratio between 0.17-0.24, collectively exhibiting low Poisson's ratio and low-to-moderate Young's modulus characteristics. This suggests favorable geological conditions for hydraulic fracturing in the Sha-1 Member, though vertical mechanical heterogeneity may increase the risk of uncontrolled fracture height growth, necessitating 3D geomechanical modeling to optimize fracture design parameters. Utilizing established rock mechanical parameter correlations and well log data, comprehensive profiles including dynamic/static Young's modulus, dynamic/

static Poisson's ratio, internal friction coefficient, and uniaxial compressive strength were established. Based on conventional logging data and imaging data, a reservoir geomechanical model is established. The model parameters are calibrated using data from drilling, formation testing, fracturing tests, and core experiments (Figure 1).

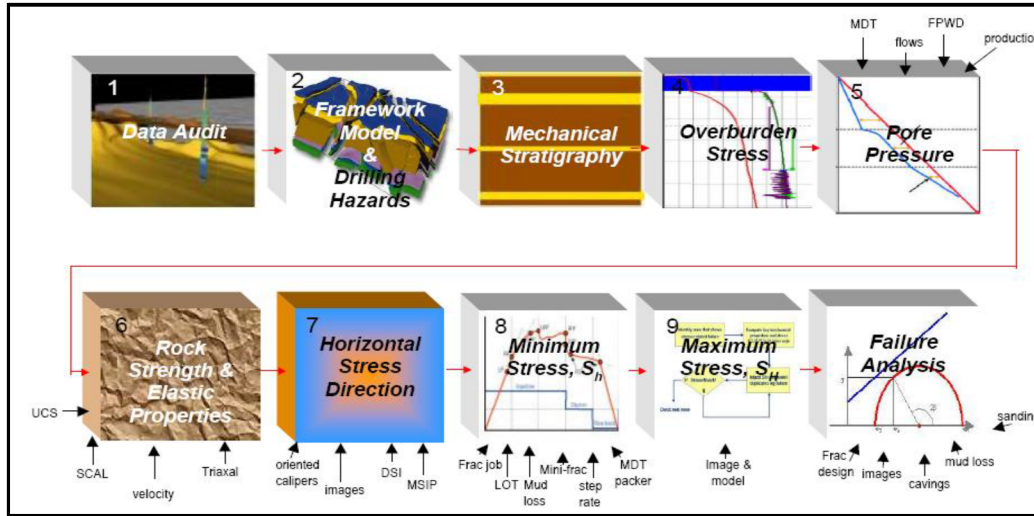


Figure 1—Geomechanical Model Workflow

Rock Mechanics Modeling

Dynamic elastic modulus is calculated using the P-wave and density logs. The elastic modulus is referred to as dynamic because the acoustic logging is conducted at high frequencies (approximately 10 kHz). Compared to mechanical property testing in laboratories or under stable loads at the surface/near the wellbore, rocks exhibit stiffer characteristics under dynamic conditions. The dynamic elastic modulus and Poisson's ratio are calculated using the following correlation:

$$E_{\text{dyn}} = \rho V_s^2 (3V_p^2 - 4V_s^2) / (V_p^2 - V_s^2) \quad (1)$$

$$\nu_{\text{dyn}} = (V_p^2 - 2V_s^2) / 2(V_p^2 - V_s^2) \quad (2)$$

Elastic Modulus. The elastic modulus E of a rock refers to the ratio of axial stress to the increment of axial strain under uniaxial compression conditions, as shown in the following equation:

$$E = \Delta \delta_z / \Delta \varepsilon_z \quad (3)$$

In this equation, E —the elastic modulus—is the slope of the stress-strain curve, representing the rate of change of stress with respect to strain under uniaxial stress conditions.

$\Delta \delta_z$, $\Delta \varepsilon_z$ —increments of axial stress and strain.

It has a significant impact on fracture width and fracturing pressure. In a two-dimensional model where the fracture height is assumed to be constant, for Newtonian fluids, the fracture width is inversely proportional to the fourth root of the elastic modulus, i.e., $W \sim 1/E^{1/4}$; for non-Newtonian fluids, the relationship between fracture width and elastic modulus is $W \sim 1/E^{1/(2n'+2)}$, where n' is the power-law index of the non-Newtonian fluid. Whether in a two-dimensional model or a three-dimensional model, calculations show that as the elastic modulus of the rock increases, the fracture width decreases.

According to the two-dimensional fracture model, for Newtonian fluids, the net pressure during fracturing operations is proportional to the 3/4 power of the elastic modulus, i.e., $\Delta p(t) \sim E^{3/4}$; for non-Newtonian fluids, the relationship between pressure and elastic modulus is $\Delta p(t) \sim E^{(2n'+1)/(2n'+2)}$. As the elastic modulus increases, the net pressure during the operation will also increase.

Poisson's ratio. The Poisson's ratio of a rock refers to the ratio of lateral strain to axial strain under uniaxial compression conditions.

$$\nu = -\frac{\epsilon_r}{\epsilon_z} \quad (4)$$

Poisson's ratio is an important parameter in determining horizontal ground stress. Generally, due to the relationship between lateral strain and axial strain, as well as the relationship between axial strain and axial stress, which are often in the form of curves, the Poisson's ratio of rock tends to increase with increasing axial stress. The linear segment of the stress-strain curve is commonly used as the basis for calculations.

The deformation modulus and Poisson's ratio of a rock mass are influenced by various factors, including the mineral composition, structure and texture of the rock, degree of weathering, porosity, moisture content, microstructural surfaces, and their relationship with the load direction, leading to significant variability.

Shear Modulus. The shear modulus is the ratio of shear stress to strain within the elastic deformation limit under shear stress, also known as the modulus of rigidity or rigidity modulus. It is one of the mechanical property indicators of a material, representing the material's ability to resist shear strain. A larger modulus indicates stronger rigidity, making the rock more resistant to shear deformation; when the shear modulus of a rock is infinite, the rock becomes a rigid body. The reciprocal of the shear modulus is called the shear compliance, which measures the amount of shear strain under a unit shear force, indicating the ease of shear deformation of the material.

The shear modulus G of a rock is related to its elastic modulus E and Poisson's ratio ν as follows:

$$G = \frac{E}{2(1+\nu)} \quad (5)$$

Rock Compressive Strength. This refers to the ultimate strength of a rock specimen under uniaxial pressure at the point of failure, numerically equal to the maximum axial stress at failure, usually denoted as σ_c :

$$\sigma_c = \frac{P}{A} \quad (6)$$

In the equation, P represents the applied load at failure, known as the failure load; A is the original cross-sectional area.

If the results from rock mechanics laboratory measurements are available, this correlation will be calibrated by matching the rock mechanical properties derived from well logging with the laboratory measurement results.

Pore Pressure Modeling

Pore Pressure Prediction Technology

Undercompaction Pore Pressure Prediction Technique. The conventional theory (Jones, 1969) for predicting abnormal formation pore pressure attributes overpressure to undercompaction caused by disequilibrium compaction in argillaceous sediments. This technique assumes that equivalent depths within the same compaction zone yield equal acoustic transit times, resulting in a pore pressure calculation model based on the normal compaction trendline:

$$P_{fa} = G_{oa}H_a - (G_0 - G_w)(\ln t_0 - \ln t)/c \quad (7)$$

Where: P_{fa} = Formation pore pressure in target zone; H_a = Burial depth of target zone (m); G_0 = Intercept term of normal trendline; t = Acoustic transit time in target zone ($\mu\text{s}/\text{m}$); c = Slope of normal trendline. This method is applicable only to overpressures caused by disequilibrium compaction-induced undercompaction. It requires shale formations within the target interval to establish the normal compaction trend equation.

Accuracy significantly decreases for formations where overpressure originates from mechanisms other than undercompaction.

Terzaghi's effective stress principle. Terzaghi's effective stress principle (Terzaghi, 1923) defines the quantitative relationship among rock effective stress, overburden pressure, and formation pore pressure:

$$\sigma = P_0 - P_p \quad (8)$$

$$P_0 = 0.00980665 \times \left(\rho \times H_0 + \int_{H_0}^H \rho_b dH \right) \quad (9)$$

Where: ρ, ρ_b = Formation average and bulk density (g/cm^3); H, H_0 = Start depth and target depth of study interval (m) Limitations: Considers only burial depth effects on pore pressure; Ignores influences of lithological variations and fluid properties; Lacks compensation for diagenetic or thermal alteration effects.

Equivalent Depth Method. This technique constructs pore pressure predictions (Hottman, 1965) using measured parameters at identical burial depths:

$$d_{eq1A} = GGA - \frac{ZB}{ZA} (GGB - d_{eq1B}) \quad (10)$$

Where: d_{eq1A}, d_{eq1B} = Formation bulk densities at equivalent depths (g/cm^3); ZB = Equivalent depth (m); GGA, GGB = Overburden pressure gradients at locations A and B (MPa/m). Limitations: 1. Solely considers burial depth effects on pore pressure; 2. Neglects influences of lithological variations and fluid properties; 3. Fails to account for diagenetic or thermobaric effects.

Bowers Method. This pore pressure prediction model (Bowers, 1989) integrates overburden pressure with seismic interval velocity:

$$P_p = S_v - b \sqrt{\frac{V - V_{mudline}}{a}} \quad (11)$$

Where: P_p = Predicted pore pressure (MPa); S_v = Overburden pressure (MPa); V = Interval velocity (m/s); $V_{mudline}$ = Shale baseline velocity (m/s).

Pore Pressure Prediction

Pore pressure evaluation determines fluid pressure within formation pores at varying depths. In drilled wells, direct measurements can be obtained using tools such as RFT (Repeat Formation Tester) or MDT (Modular Formation Dynamics Tester), or through well testing. While providing reliable spot data, these methods yield sparse measurements without continuous profiles. In practice, pore pressure is primarily calculated via normal compaction trendline methods. These techniques quantitatively determine pore pressure by analyzing deviations of well log data from normal compaction trends using established models. The Eaton method (Eaton, 1975) and Equivalent Depth method represent the most widely applied approaches. Eaton method calculates formation pressure using normal compaction trendlines. It employs a depth-independent power-law relationship between pore pressure and acoustic transit time:

$$P_p = P_{ob} - (P_{ob} - P_h) \left(\frac{\Delta t_n}{\Delta t_o} \right)^N \quad (12)$$

Where: P_p = formation pore pressure; P_{ob} = overburden pressure; P_h = hydrostatic pressure; Δt_n = normal shale transit time at depth; Δt_o = measured shale transit time; N = Eaton exponent (formation-dependent coefficient). This technique applies exclusively to shale formations, with the N exponent varying by region and geological age (e.g., 3.0 for Gulf of Mexico). Eaton (1976) modified the method as follows:

$$P_p = P_{ob} - (P_{ob} - P_h) \left(\frac{V}{V_n} \right)^3 \quad (13)$$

Where: V_n = velocity in normally compacted formation; V = measured formation velocity.

In this study, Eaton's theory was primarily employed for pore pressure prediction across wells based on formation overpressure mechanisms in the research area. Well log data were utilized for calibration to establish reliable model parameters, subsequently applied in 1D geomechanical modeling.

Three of six wells in this geomechanical modeling study contained pressure test data within the Sha-1 Member (Wells Jinqian 201, Jinqian 205, and Jinqian 205-2-H1). Supplementary pressure tests from six offset wells were incorporated for pore pressure model calibration. The integrated pore pressure model combines well logs, compaction trends, drilling mud weights, and measured pressure data (Table 1).

Table 1—Formation Pressure Data of typical wells

Well	Jinqian201	Jinqian203	Jinqian205	Jinqian206	Jinqian207	Jinqian208	Jinqian209	Jinqian19			Jinqian219H	
Formation	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	Sha-1	
Test Method	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	Perf & Sand	
Test Interval (m)	2312.00-3858.00	2173.00-2302.00	2572.00-3617.00	2515.00-3608.00	2426.00-2483.00	2569.00-2696.00	2191.00-2229.00	2672.00-2705.00	2576.00-2608.00	2672.00-2705.00-2576.00-2608.00	1940.00-3308.00	
Stimulation Method	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	Sand Frac	
Test Date (YYYY.MM.DD)	2024.9.28	2024.7.9	2024.9.12	2024.10.9	2024.10.9	2024.11.21	2024.9.26	2023.10.10	2023.10.24	2023.11.14	2024.8.14	
Stabilization	Time (h:min)	4:00	6:00	4:00	4:00	5:00	6:40	6:00	8:00	4:00	6:00	4:00
	Casing Pressure (MPa)	8.73	7.49	12.82	13.2	8.59	8.43	11.04	8.79	12.572	11.719	6.64
Fluid Level Depth(m)		1873.79						2057.2				
Flowing Pressure (MPa)		11.622	9.91	16.37	17.837	12.1792			18.51	16.59	15.54	9.235
	Gas (104m3)	13	4.66	30.06	41.2668	9.28	9.74	2.1	1.10	22.15	24.25	24.77
	Open-flow Potential(104m3)	19.84	7.05	55.99	94.6	12.24	13.59	3.28	2.68	43.83	41.86	37.13
Max	Casing Pressure (MPa)	11.6	10.77	17.07	16.33	15.66	15.78	15.7	10.14	15.88	15.54	11.21
Formation Pressure (MPa)		16.21	13.923	20.31	20.637	19.7712	19.24	20.513	21.08	20.09	20.00	13.057
Formation Temperature (°C)		69.52	69.79	68.65	70.94	71.87	71.40	69.46		69.33	74.17	62.72
Conclusion		Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone	Gas Zone
Remarks		No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S	No H ₂ S
Pressure Coefficient		0.78	0.64	0.91	0.9125	0.9	0.87	0.96	0.88	0.87	0.85	0.77
Tested Sand Group		1	1、2	2	2	2	1、2	1、2	1	2	1、2	1

In Well Jinqian 201, the target Sha-1 Member exhibits pore pressures of 11.8-21.7 MPa (pressure gradient 0.63-0.89 SG). Well Jinqian 205 shows 14.3-22.2 MPa pore pressure (0.67-0.95 SG gradient) in the same interval. Identical methodology was applied for pore pressure interpretation in other wells (Figure 2).

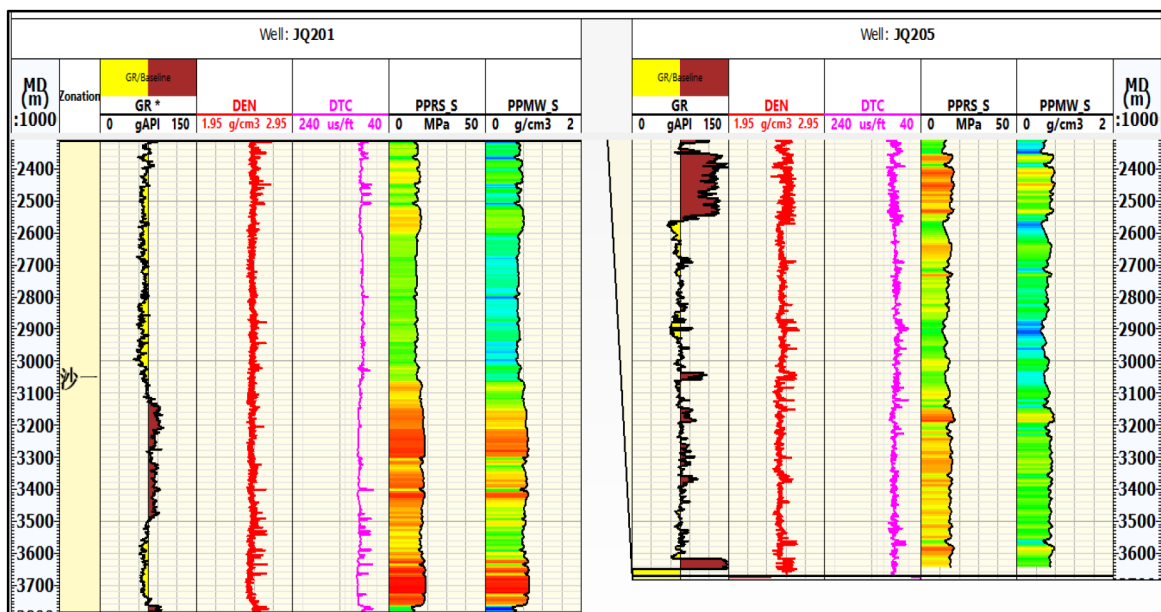


Figure 2—Single-Well Pore Pressure Analysis of Typical Well

The pore pressure model integrates multi-source data using Jinqian 201 and 205 as representative wells. Key workflow components include: 1) Log response analysis: Establishing distinct compaction trends for sand-shale sequences via acoustic/density logs, with lithology-based segmentation eliminating formation discontinuity effects; 2) Drilling dynamics inversion: Calibrating pressure responses in sensitive zones using offset well events (kicks/losses) and mud weight adjustments; 3) Measured pressure constraints: Incorporating Modular Formation Tester (MDT) data to validate pressure system distribution. The Sha-1 Member exhibits characteristic "overpressure-underpressure-overpressure" oscillations, where interbedded shales create regional pressure compartments driven by undercompaction and hydrocarbon generation, while permeable sandstones develop natural pressure-release pathways.

Horizontal Stress Modeling

Stress Orientation Analysis

Borehole breakouts represent wellbore intervals experiencing shear failure. During drilling, mud replacement redistributes in-situ stresses, creating localized stress concentrations. In vertical wells, maximum circumferential compressive stress occurs along the S_{hmin} direction. When this stress exceeds formation compressive strength, shear failure causes rock spallation, forming an elliptical borehole with its major axis (breakout orientation) parallel to S_{hmin} . In regions where overburden stress dominates, continuous breakouts develop vertically symmetric features along the minimum horizontal stress direction. Borehole shear failure occurs when circumferential compressive stress exceeds formation strength, causing elliptical boreholes with major axes (breakout orientation) parallel to S_{hmin} and vertically symmetric features in vertical wells. Observed instability in offset Well Qianshan 203 -9-H1 ([Figure 3](#)) indicates a NW-SE maximum horizontal stress orientation at $\approx 110^\circ$.

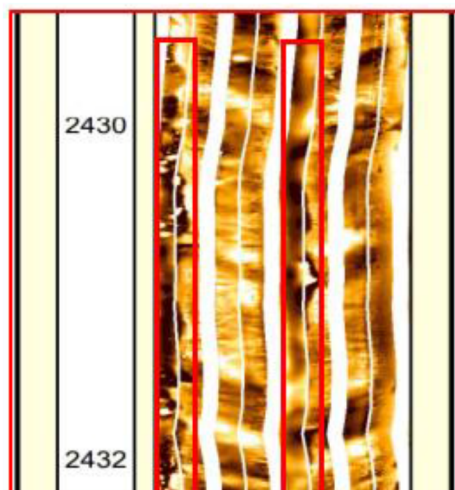


Figure 3—Borehole Breakout Characteristics in Reservoir Section,

Microseismic monitoring of Well Jinqian 201's hydraulic fracturing recorded fracture orientations across 13 stages indicating a NW-SE ($\approx 108^\circ$) maximum horizontal stress direction consistent with regional trends. Borehole breakout inversion applies Coulomb Fault Theory (Figure 4), where distinct stress regimes correspond to specific S_v - S_{hmin} - S_{hmax} relationships (Brace, 1980): Normal Faulting (NF: $S_{hmin} < S_{hmax} < S_v$), Strike-Slip (SS: $S_{hmin} < S_v < S_{hmax}$), and Reverse Faulting (RF: $S_v < S_{hmin} < S_{hmax}$). Polygonal boundaries represent frictional equilibrium states common in crustal formations. The theory assumes fault slip occurs when shear-to-effective-normal stress ratio exceeds 0.6 friction coefficient (Byerlee, 1978). S_{hmax} magnitude is efficiently constrained using borehole rock strength or observed breakout dimensions, with red contours indicating feasible S_{hmax} ranges derived from Unconfined Compressive Strength (UCS) and breakout width at given depths.

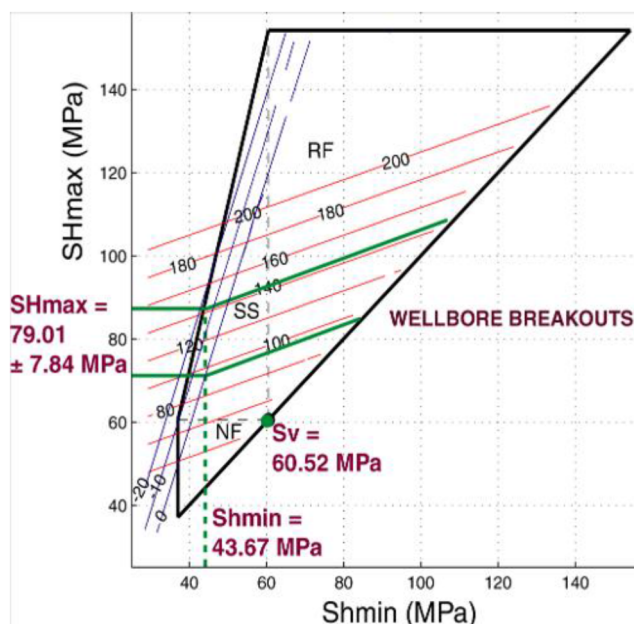


Figure 4—Simulation Methodology for Maximum Horizontal Stress (S_{hmax})

S_{hmin} and S_{hmax} magnitudes derived from well logs are estimated using a poroelastic model:

$$S_{hmin} = \frac{\nu}{1-\nu} S_V + \frac{1-2\nu}{1-\nu} \alpha P_p + \frac{E}{1-\nu^2} \varepsilon_h + \frac{\nu E}{1-\nu^2} \varepsilon_H$$

$$S_{Hmax} = \frac{\nu}{1-\nu} S_V + \frac{1-2\nu}{1-\nu} \alpha P_p + \frac{E}{1-\nu^2} \varepsilon_H + \frac{\nu E}{1-\nu^2} \varepsilon_h$$

Where: ν = static Poisson's ratio; α = Biot's coefficient; P_p = pore pressure; E = static Young's modulus; S_V = vertical stress; ε_h , ε_H = tectonic strains in S_{hmin} and S_{Hmax} directions. The final stress model integrates effective stress ratios with log-based calculations. While Biot's coefficient is uniformly 0.9 regionally, tectonic stress coefficients vary across wells due to structural positioning and reservoir heterogeneity. Modeling indicates a dominant strike-slip stress regime ($S_{Hmax} > S_V > S_{hmin}$), favoring vertical fracture development during reservoir stimulation.

In the Sha-1 Member of Well Jinqian 201, minimum horizontal principal stress ranges from 37-45 MPa and maximum horizontal principal stress from 45-62 MPa. The target zone exhibits moderate-to-low horizontal stress differential (9-18 MPa), facilitating complex branched fracture networks. Elevated net fracture pressure overcomes natural fracture activation thresholds, enabling effective network stimulation. As shown in Figure 5, poroelastic models were applied to establish horizontal stress profiles for all wells based on rock mechanics interpretations.

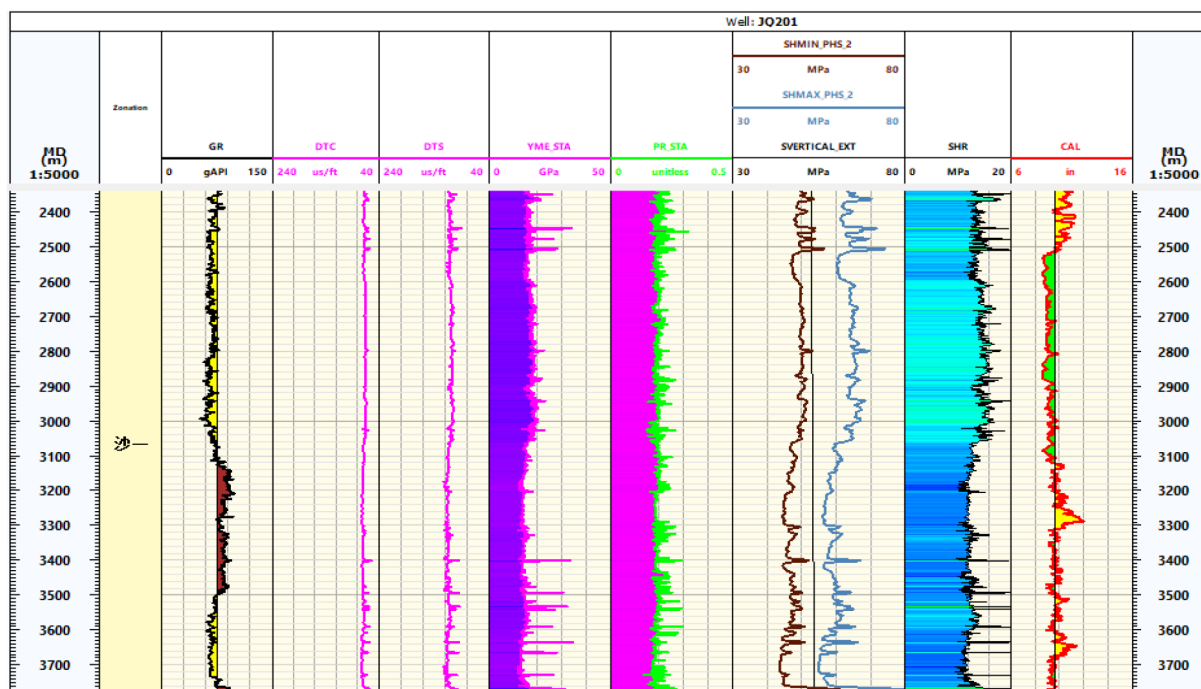


Figure 5—Horizontal Stress Interpretation Model for Jinqian 201 Well

3D Geomechanical Modeling Results

Building upon 1D geomechanical analysis, a 3D geomechanical model incorporating structural frameworks and seismic data accounts for reservoir spatial anisotropy. This model quantifies maximum/minimum principal stresses, overburden pressure, horizontal stress differential, Young's modulus, Poisson's ratio, and unconfined compressive strength. Laboratory rock mechanics testing—including mechanical parameter validation, velocity anisotropy analysis, paleomagnetic orientation, and triaxial compression experiments—provides stress magnitude/orientation, gradients, and rock properties critical for stimulation design. Integrated with well logs, lithology distributions, pore pressure, and fluid properties, these datasets enable comprehensive 3D characterization of reservoir geomechanical behavior through structural horizon modeling, fault distribution analysis, and seismic interpretation.

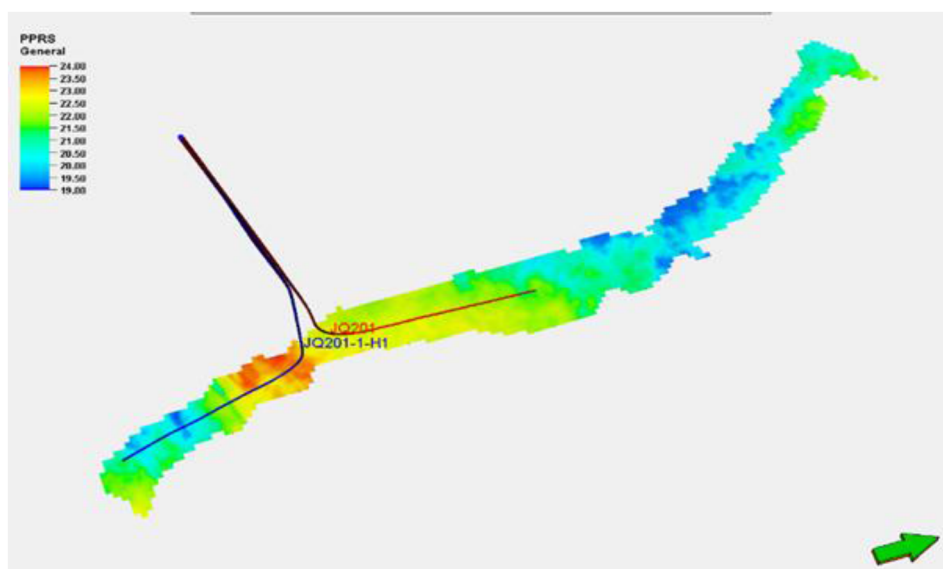


Figure 6—3D Geomechanical Model (Pore Pressure)

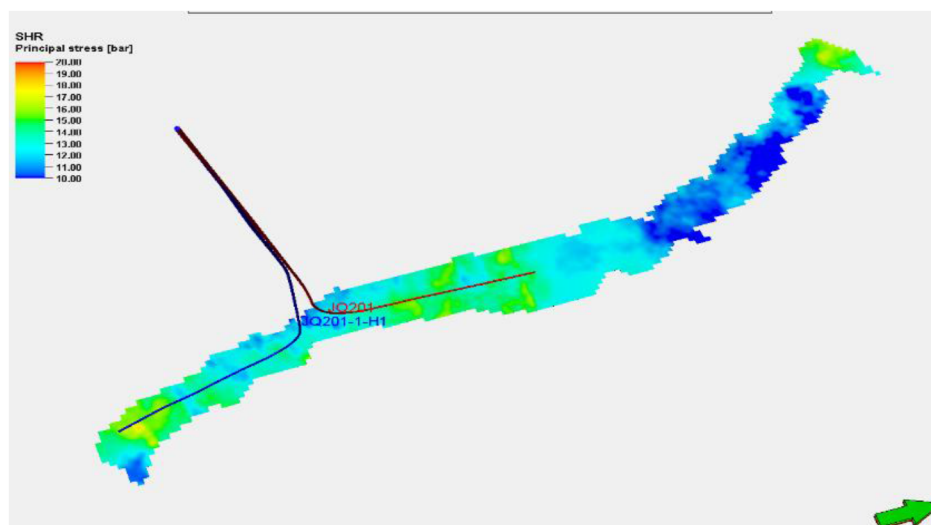


Figure 7—3D Geomechanical Model (Differential Horizontal Stress)

Optimization of Fracturing Parameter

Reasonable well spacing is the result of comprehensive consideration of geological and engineering factors, such as improving inter-well reserves mobilization, reducing inter-well interference, and lowering development costs. It is also a key focus of research for the economic development of shale oil. The shale reservoir in the Lu-1 Formation is typically developed using small well spacing in single-layer platform wells or multi-layer well patterns. In this study, different well spacings of 100 m, 200 m, and 300 m were set for both single-layer platforms and multi-layer well patterns. Zipper fracturing was employed to simulate fracture propagation and stress field changes under different well spacings. The results show that for a single-layer platform with a well spacing of 100 m, severe inter-well stress interference occurs, leading to shorter vertical fracture extension, localized fracture overlap, and uneven fracture network propagation. The excessive overlap results in a high degree of repeated reservoir stimulation, reaching 31.32%. Vertically, fractures from two wells interconnect, increasing the risk of casing damage in adjacent wells. When the well spacing is increased to 200 m, inter-well stress interference is significantly reduced, and its impact on fracture propagation is also minimized. The degree of repeated stimulation decreases to 13.71%. At

a well spacing of 300 m, inter-well stress interference essentially disappears, and stress has almost no noticeable effect on fracture propagation. Fractures propagate in a balanced and symmetrical manner, with no significant overlap between wells. Compared to single-layer platforms, inter-well interference is more severe in multi-layer well patterns. At a well spacing of 100 m, significant stress interference occurs between wells, with stress superposition from the outer wells onto the central well exceeding 6 MPa. Fractures propagate unevenly between wells, limiting vertical fracture length and directly connecting multiple clusters to adjacent wells, thereby increasing the risk of casing deformation in neighboring wells. When the well spacing is increased to 200 m, inter-well stress interference is significantly reduced, and artificial fractures achieve full coverage. Vertical fracture extension becomes more extensive with increased well spacing, indicating that larger well spacing helps reduce inter-well flow and stress interference. When the well spacing is further increased to 300 m, inter-well stress interference essentially disappears, and fractures propagate evenly between wells, forming a well-structured fracture network. However, a significant portion of the reservoir between wells remains outside the stress influence range, reducing reservoir stimulation and reserves mobilization (see Figure 8).

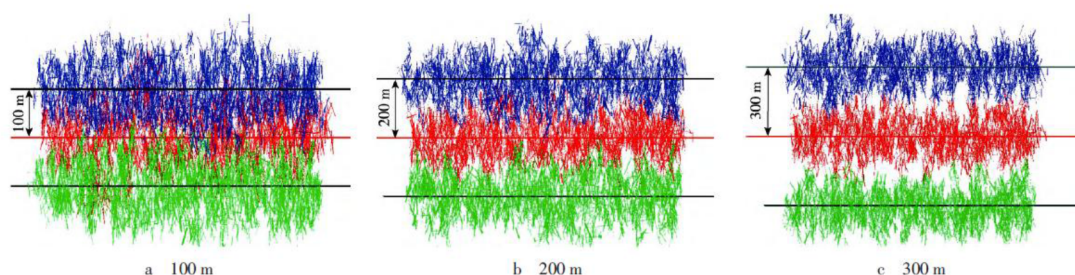


Figure 8—Inter-Well Fracture Propagation Under Different Well Spacings in a Multi-Layer Well Pattern

Fracturing simulations indicate that proppant volume, cluster spacing, and well spacing are key factors influencing the effectiveness of reservoir fracturing. While a larger proppant volume can rapidly expand the stress-affected area, increasing the stimulated reservoir volume (SRV) as well as the number and length of fractures, exceeding 1,600 m³ of proppant can activate natural fractures in the reservoir. This leads to fracturing fluid loss along these natural fractures, slowing the growth rate of single-cluster stimulation or causing repeated stimulation, ultimately degrading the fracturing results. A similar relationship exists between cluster spacing, well spacing, and fracturing effectiveness. When the cluster spacing and well spacing are less than 6 m and 200 m, respectively, reducing these distances enhances single-cluster reservoir stimulation and improves reserves mobilization. Simulation results also show balanced inter-well fracture propagation. However, as cluster spacing and well spacing are further reduced, inter-cluster and inter-well interference intensifies, leading to repeated reservoir stimulation and poorer fracturing outcomes. Additionally, the half-length of fractures can reach up to 105 m. When the well spacing is less than 200 m, excessively long fractures can easily damage the casing of adjacent wells, increasing the risk of casing deformation. Therefore, considering both reservoir stimulation effectiveness and balanced fracture propagation, the optimal proppant volume is 1,600 m³, the reasonable cluster spacing is 5–7 m, and the appropriate well spacing for single-layer platforms and multi-layer well patterns is 200–250 m.

Conclusions

The channel area of Jinqian 201 Platform exhibits robust formation energy (mean pore pressure: 21 MPa), enhancing tight gas flow and recovery. Hydraulic fracturing replenishes reservoir energy, providing essential post-stimulation productivity assurance. Uniform stress distribution features minimum horizontal stress averaging 40 MPa. Fracture initiation pressure demonstrates approximate proportionality to minimum horizontal stress, particle normal contact stiffness, and bond strength. This moderate stress

regime facilitates stable fracture propagation—avoiding both prohibitive high-pressure constraints and uncontrolled low-pressure growth. Moderate-to-low horizontal stress differential (mean $\Delta\sigma=11$ MPa) critically influences fracture geometry. Defined as $SH_{max} - SH_{min}$, this differential governs fracture orientation and morphology within the formation.

Fracturing operations employed hybrid slickwater/gel fluids with configured fluid-loss coefficients, viscosity, and friction. Composite proppants were selected with closure-pressure-dependent conductivity and compressive strength. Complex fracture network simulations integrated actual well trajectories, treatment data, stage/cluster designs, 3D geomechanics, and natural fracture models. Considering both reservoir stimulation effectiveness and balanced fracture propagation, the optimal proppant volume was determined to be 1,600 m³, with a cluster spacing of 5–7 m and a well spacing of 200–250 m for both single-layer platforms and multi-layer well patterns. Three types of fracture networks were formed post-fracturing in the reservoir. The study area exhibited a "hydraulic main fracture" type network, with post-fracturing half-lengths ranging from 70–105 m and fracture heights of 20–30 m. Although the "hydraulic main fracture" type fractures exhibited greater half-lengths and heights, leading to more thorough reservoir stimulation, reducing well spacing and cluster spacing can easily result in inter-well channeling and casing deformation, ultimately leading to a loss in final recovery efficiency.

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